Taehyung Noh

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Research Interests

My research interest lies in Human-AI Interaction, with a focus on developing trustworthy and personalized AI systems.

Experience

- Hanyang Human-Centered Computing Laboratory [] — *Jan. 2022 - Present*
  Research Assistant — Seoul, Korea

- Ajou Human-Computer-Interaction Lab at Dept. of Software — *Aug. 2020 - Dec. 2021*
  Undergraduate Research Assistant — Suwon, Korea

Education

- Hanyang University — *Mar. 2022 - Present*
  M.S. & Ph.D. Integrated Student, Department of Artificial Intelligence (Advisor: Kyungsik Han) — Seoul, Korea
- Ajou University — *Mar. 2018 - Feb. 2022*
  Bachelor of Engineering, Department of Software — Suwon, Korea

Projects

- KETI Industrial AI Data Preprocessing Platform
  Present
  - Developing an AI data preprocessing platform for industrial applications.
  - Providing an integrated solution for efficiently processing and analyzing data across various industrial domains.

- PALETTE: Value-Based User Understanding (CIKM)
  First Author — 2026
  - PALETTE is a theory-driven benchmark, grounded in Schwartz's Theory of Basic Human Values, for evaluating whether LLMs can infer a user's latent values from behavioral cues and transfer them to new contexts.
  - Constructed 25,200 dialogues across 400 personas spanning two tasks (Dialogue-Based Understanding and New-Context Application), revealing that LLMs generate appropriate responses without truly comprehending the underlying values.

- LLM Deceptive Persuasion Study (CHI)
  2026
  - Developed a taxonomy of LLM deceptive strategies and conducted a large-scale empirical study on whether LLMs can persuade humans through deception.
  - Investigated how different deceptive strategies affect human decision-making in human-AI interaction.

- TRIPLE (AAAI)
  First Author — 2025
  - TRIPLE (Theory-guided Reasoning for Intent and habIt Profiling with LLMs for pErsonalization) is a novel framework that integrates dual-process theory and the Theory of Planned Behavior (TPB) into LLM-based user modeling.
  - Constructs both habitual and intentional behavior profiles, then generates behavioral rationale that explains the interaction between these processes to predict user behavior.

- TRIPLE (CIKM)
  First Author — 2025
  - TRIPLE is a profiling technology that combines the Theory of Planned Behavior (TPB) with Large Language Models (LLMs).
  - Uses LLMs to understand a user's psychological motivations and refines their profile by comparing predictions with actual behavior, dramatically improving personalization services.

- Red Teaming Attack Dataset Analysis
  Project Leader — 2024

- Analyzed real-world LLM vulnerabilities using attack data from the 2023 DEF CON 31 Generative AI Red Teaming Challenge.
- Preprocessed and relabeled the dataset to identify which target categories are most susceptible to harmful-content attacks.
- Classified victim groups and categorized prompt types to reveal systematic patterns in successful jailbreaks.

- Multi-Agent Personality Detection System (PADO)
  Second Author 2024
  - PADO is a multi-agent system that detects personality traits (OCEAN) from user-generated text.
  - Multiple specialized agents collaborate to perform more accurate personality analysis.

- Fashion-FINE
  Third Author 2024
  - Fashion-FINE is a Vision-Language Pre-training model for fine-grained fashion retrieval.
  - Introduces modality-agnostic adapter, hard negative mining with focal loss, and comprehensive cross-modal alignment.

- My Own Style (MOS)
  First Author May. 2022 - 2025
  - Investigated how domain expertise shapes user perception and satisfaction with fashion-recommendation outcomes.
  - Built MOS dashboard that analyzes outfit images and recommends similar/diverse fashion items.
  - Large-scale study with 166 participants showing fashion experts prefer highly similar recommendations while non-experts prefer diverse results.

Publications

Conference Papers

[C.1] Taehyung Noh et al. (2026). From Preferences to Values: Evaluating Latent User Understanding and Transfer in LLMs. The ACM International Conference on Information and Knowledge Management (CIKM 2026 Short, Under Review)

[C.2] Haein Yeo, Seungwan Jin, Taehyung Noh, Yejin Shin, Sangyeon Kang, Sangwoo Heo, Jiwon Chung, Hwarim Hyun, Kyungsik Han (2026). "Can LLMs Persuade Humans with Deception?": From a Deceptive Strategy Taxonomy to a Large-Scale Empirical Study. The ACM Conference on Human Factors in Computing Systems (CHI 2026)

[C.3] Taehyung Noh, Seungwan Jin, Haein Yeo, Kyungsik Han (2026). TRIPLE: Theory-Driven Integration of Planned and Habitual Behaviors for LLM-based Personalization. The AAAI Conference on Artificial Intelligence (AAAI 2026, Oral, Acceptance Rate: 17.6%)

[C.4] Taehyung Noh, Seungwan Jin, Haein Yeo, Kyungsik Han (2025). Externalizing Social-Cognitive Structures for User Modeling: Toward Theory-Driven Profiling with LLMs. The ACM International Conference on Information and Knowledge Management (CIKM 2025 Short, Acceptance Rate: 27.2%)

[C.5] Haein Yeo, Taehyung Noh, Seungwan Jin, Kyungsik Han (2025). PADO: Personality-induced multi-Agents for Detecting OCEAN in human-generated texts. The International Conference on Computational Linguistics (COLING 2025)

[C.6] Seungwan Jin, Hoyoung Choi, Taehyung Noh, Kyungsik Han (2024). Integration of global and local representations for fine-grained cross-modal alignment. The European Conference on Computer Vision (ECCV 2024)

[C.7] Taehyung Noh, Haein Yeo, Myungin Kim, Kyungsik Han (2023). A study on user perception and experience differences in recommendation results by domain expertise: the case of fashion domains. The ACM Conference on Human Factors in Computing Systems Late-Breaking Work (CHI LBW 2023, Acceptance Rate: 36%)

Journal Papers

[J.1] Haein Yeo, Taehyung Noh, Kyungsik Han (2025). LLM-Generated Content-Based Explanations for User Experience in Fashion Recommender Systems. Fashion and Textiles

Technical Reports

[R.1] Yeajin Shin, Kyungsik Han, Taehyung Noh, Mingon Jeong (2025). LLM Attack Strategies. TTA Report

Patents

[P.1] Seungwan Jin, Kyungsik Han, Taehyung Noh, Junghyun Kim (2025). Dynamic Persona-based Prediction System and Method for Personalized Long-term Mental Health Monitoring.

[P.2] Kyungsik Han, Taehyung Noh (2025). Method and Device for Generating User Profiles Using Artificial Intelligence.

[P.3] Bogoan Kim, Dayoung Jung, Taehyung Noh, Kyungsik Han (2023). Visualization Analysis Tool Equipped with VR Multimodal Sensor Data Processing Pipeline Collected from People with Autism Spectrum Disorder.

[P.4] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2023). Device and Method for Providing Dashboard Services on Fashion Image Analysis.

[P.5] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2023). Device and Method for Similar Image Recommendation Using Fashion Attributes and Image.

[P.6] Kanghwi Ahn, Junseok Ryu, Kyungsik Han, Taehyung Noh (2023). Method for Analyzing Image and Text-based Characteristics of Objects.

[P.7] Taehyung Noh, Kyungsik Han (2026). Method and Device for Generating User Profiles Using Artificial Intelligence. (PCT International Application, PCT/KR2026/095009)

[P.8] Kyungsik Han, Seungwan Jin, Hoyoung Choi, Taehyung Noh (2022). Image-Text Retrieval Model Construction Method and Service Device.

Software Registrations

[SW.1] Kyungsik Han, Taehyung Noh (2025). Human-in-the-Loop Industrial Preprocessing & Training System (HILIPS). (C-2025-052302)

[SW.2] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2025). Customized Fashion E-commerce Solutions Dashboard. (C-2025-025185)

[SW.3] Kyungsik Han, Bogoan Kim, Dayoung Jeong, Taehyung Noh (2024). VR Data-based Expert Decision-making and Patient Self-cognition Support Dashboard for Depression Patients. (C-2024-034698)

[SW.4] Kyungsik Han, Taehyung Noh, Haein Yeo, Myungjin Kim (2024). Fashion Element Analysis and Product Description Generation Dashboard for Fashion Experts. (C-2024-016124)

[SW.5] Taehyung Noh, Kyungsik Han (2022). VR User Biometric Signal Data Collection and Analysis System. (C-2022-051352)

Honors and Awards

Best Paper Award, Korea Data Mining Society	Nov. 2024
Best Presentation Award, Korea Software Congress (KSC 2023)	Dec. 2023
Best Inventor Award, Seoul International Invention Fair	Dec. 2022
Participation Award, mySUNI Creative Challenge	Dec. 2022

TEACHING EXPERIENCE

Co-lecturer

- ITE3062: Human-Computer Interaction (In English) — Fall 2024

Teaching Assistant (TA)

- SOI3010: Laboratory practice of Intelligence Computing 2 — Fall 2024

ACADEMIC SERVICES

Conference Reviewer

- Conference on Empirical Methods in Natural Language Processing (EMNLP) — 2024
- ACM Conference on Human Factors in Computing Systems (CHI) — 2024
- ACM International Conference on Information and Knowledge Management (CIKM) — 2023

References

1. Kyungsik Han
   Associate Professor, Hanyang University
   Department of Artificial Intelligence
   kyungsikhan@hanyang.ac.kr